```python
import re
import string
import nltk
from nltk.corpus import brown, reuters, stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import PorterStemmer, WordNetLemmatizer
from sklearn.datasets import fetch_20newsgroups
import spacy
```

```python
nltk.download('brown')
nltk.download('reuters')
nltk.download('punkt_tab')
nltk.download('stopwords')
nltk.download('wordnet')
```

```
[nltk_data] Downloading package brown to /root/nltk_data...
[nltk_data]   Package brown is already up-to-date!
[nltk_data] Downloading package reuters to /root/nltk_data...
[nltk_data]   Package reuters is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
True
```

```python
stop_words = set(stopwords.words('english'))
stemmer = PorterStemmer()
lemmatizer = WordNetLemmatizer()

def normalize(text):
    return text.lower()

def clean_text(text):
    text = re.sub(r'\d+', '', text)
    text = text.translate(str.maketrans('', '', string.punctuation))
    return text

def tokenize(text):
    return word_tokenize(text)

def remove_stopwords(tokens):
    return [word for word in tokens if word not in stop_words]

def stem_tokens(tokens):
    return [stemmer.stem(word) for word in tokens]

def lemmatize_tokens(tokens):
    return [lemmatizer.lemmatize(word) for word in tokens]

def preprocess(text):
    text = normalize(text)
    text = clean_text(text)
    tokens = tokenize(text)
    tokens = remove_stopwords(tokens)
    stems = stem_tokens(tokens)
    lemmas = lemmatize_tokens(tokens)
    return tokens, stems, lemmas
```

```python
brown_text = " ".join(brown.words()[:500])
tokens, stems, lemmas = preprocess(brown_text)

print("Tokens:", tokens[:20])
print("Stems:", stems[:20])
print("Lemmas:", lemmas[:20])
```

```
Tokens: ['fulton', 'county', 'grand', 'jury', 'said', 'friday', 'investigation', 'atlantas', 'recent', 'primary', 'election', 'produced',
Stems: ['fulton', 'counti', 'grand', 'juri', 'said', 'friday', 'investig', 'atlanta', 'recent', 'primari', 'elect', 'produc', 'evid', 'irr
Lemmas: ['fulton', 'county', 'grand', 'jury', 'said', 'friday', 'investigation', 'atlanta', 'recent', 'primary', 'election', 'produced', '
```

```python
reuters_text = " ".join(reuters.words()[:500])
tokens, stems, lemmas = preprocess(reuters_text)

print("Tokens:", tokens[:20])
print("Stems:", stems[:20])
print("Lemmas:", lemmas[:20])
```

```
Tokens: ['asian', 'exporters', 'fear', 'damage', 'u', 'japan', 'rift', 'mounting', 'trade', 'friction', 'u', 'japan', 'raised', 'fears', '
Stems: ['asian', 'export', 'fear', 'damag', 'u', 'japan', 'rift', 'mount', 'trade', 'friction', 'u', 'japan', 'rais', 'fear', 'among', 'ma
Lemmas: ['asian', 'exporter', 'fear', 'damage', 'u', 'japan', 'rift', 'mounting', 'trade', 'friction', 'u', 'japan', 'raised', 'fear', 'amo
```

```
newsgroups = fetch_20newsgroups(subset='train')
sample_text = newsgroups.data[0]

tokens, stems, lemmas = preprocess(sample_text)

print("Tokens:", tokens[:20])
print("Stems:", stems[:20])
print("Lemmas:", lemmas[:20])
```

```
Tokens: ['lerxstwamumdedu', 'wheres', 'thing', 'subject', 'car', 'nntppostinghost', 'racwamumdedu', 'organization', 'university', 'marylanc
Stems: ['lerxstwamumdedu', 'where', 'thing', 'subject', 'car', 'nntppostinghost', 'racwamumdedu', 'organ', 'univers', 'maryland', 'colleg'
Lemmas: ['lerxstwamumdedu', 'wheres', 'thing', 'subject', 'car', 'nntppostinghost', 'racwamumdedu', 'organization', 'university', 'marylanc
```

```
nlp = spacy.load("en_core_web_sm")

text = "The quick brown foxes are jumping over the lazy dogs."
doc = nlp(text)

tokens = [token.text.lower() for token in doc if not token.is_punct]
lemmas = [token.lemma_ for token in doc if not token.is_punct]
no_stopwords = [token.text for token in doc if not token.is_stop and not token.is_punct]

print("Tokens:", tokens)
print("Lemmas:", lemmas)
print("Without Stopwords:", no_stopwords)
```

```
Tokens: ['the', 'quick', 'brown', 'foxes', 'are', 'jumping', 'over', 'the', 'lazy', 'dogs']
Lemmas: ['the', 'quick', 'brown', 'fox', 'be', 'jump', 'over', 'the', 'lazy', 'dog']
Without Stopwords: ['quick', 'brown', 'foxes', 'jumping', 'lazy', 'dogs']
```